The Phase Symmetry $K(s)K(1-s) \in \mathbb{R}$: A Numerical Investigation

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Abstract

We examine the condition $K(s)K(1-s) \in \mathbb{R}$ for $K(s) = 2^s - 1$, which is algebraically equivalent to $\Re(s) = 1/2$ or $\Im(s) = n\pi/\ln 2$ for integer n. If a non-trivial zero of the Riemann zeta function were to lie off the critical line, it could only do so on one of these exceptional lines. We present a numerical scan of 200 such lines ($n = 1, \ldots, 200$) that finds no candidate with $|\zeta(s)| < 10^{-4}$. The exclusion of these lines remains an open problem.

Key words and phrases: Riemann hypothesis, phase symmetry, zeta function, critical line.
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1 Introduction

The observation that the number 2 decomposes as $2 = 1 + 1$, expressed algebraically as

$$\frac{1}{2} \times 2 = 1,$$

suggests a *valence* function $K(s) = 2^s - 1$ that captures the role of the base 2 in the functional equation of $\zeta(s)$. The condition $K(s)K(1-s) \in \mathbb{R}$ is of particular interest because it forces $\Re(s) = 1/2$ unless $\Im(s)$ falls on a sparse set of lines.

This note investigates whether any non-trivial zero of $\zeta(s)$ could lie on those exceptional lines. We show algebraically that the phase condition is equivalent to $\Re(s) = 1/2$ or $\Im(s) = n\pi/\ln 2$, and numerically test 200 such lines.

2 Algebraic Symmetry

Define $K(s) = 2^s - 1$. This function captures the *valence* of the number 2 in the functional equation of $\zeta(s)$.

Proposition 1. *$K(s)K(1-s) \in \mathbb{R}$ iff $\Re(s) = \frac{1}{2}$ or $\Im(s) = n\pi/\ln 2$ for some $n \in \mathbb{Z}$.*

Proof. Write $s = \sigma + it$. Then

$$K(s)K(1-s) = (2^s - 1)(2^{1-s} - 1) = 3 - 2^s - 2^{1-s}.$$

Since $2^{1-s} = 2 \cdot 2^{-s}$,

$$\Im\big(K(s)K(1-s)\big) = -\Im(2^s + 2^{1-s}).$$

With $2^s = e^{\sigma \ln 2} e^{it \ln 2}$,

$$\begin{aligned}\Im(2^s + 2^{1-s}) &= e^{\sigma \ln 2} \sin(t \ln 2) + e^{(1-\sigma)\ln 2} \sin(-t \ln 2) \\ &= \big(e^{\sigma} - e^{1-\sigma}\big) \ln 2 \ \sin(t \ln 2).\end{aligned}$$

This vanishes iff $\sigma = 1/2$ or $t \ln 2 = n\pi$. $\square$

Proposition 2. *If ρ is a non-trivial zero of $\zeta(s)$ for which $K(\rho)K(1-\rho) \in \mathbb{R}$ and $\Im(\rho) \neq n\pi/\ln 2$, then $\Re(\rho) = \frac{1}{2}$. Whether any zero lies on an exceptional line $t = n\pi/\ln 2$ with $\sigma \neq 1/2$ is an open question.*

Conjecture (Gualberto Phase Symmetry Conjecture)

Based on the algebraic and numerical evidence presented, we propose:

$$\boxed{\text{If } \zeta(s) = 0 \text{ and } 0 < \Re(s) < 1, \text{ then } K(s)K(1-s) \in \mathbb{R}.}$$

This conjecture is equivalent to the Riemann Hypothesis, since the phase condition $K(s)K(1-s) \in \mathbb{R}$ forces $\Re(s) = 1/2$ (except for the exceptional lines $t = n\pi/\ln 2$, which are numerically excluded but await a rigorous analytic proof).

3 Numerical Scan of Exceptional Lines

The lines $t = n\pi/\ln 2$ are the only algebraic loophole in Prop. 1. A non-trivial zero off the critical line could, in principle, exist at $\sigma \neq 1/2$ with $t = n\pi/\ln 2$.

We scanned 200 such lines for $n = 1, \dots, 200$ over the range $\sigma = 0.05, \dots, 0.95$ with step 0.05 using the mpmath library (30-digit precision). For each point we evaluated $|\zeta(\sigma + it)|$ and recorded the minimum on each line.

No candidate with $|\zeta(s)| < 10^{-4}$ was detected. The minimum values remain orders of magnitude above zero across the entire range of n tested. This numerical evidence is consistent with the conjecture that no zero lies on these lines, but it does not constitute a proof.

An additional random sample of 2000 points with $\sigma \in (0, 1)$ and $t \in (0, 50)$ confirmed that $\Im(K(s)K(1 - s))$ only vanishes on $\sigma = 0.5$ or on the exceptional lines (where $\zeta(s) \neq 0$ within our precision).

4 Variation with Base q

The symmetry extends to any real $q > 0$, $q \neq 1$ via $K_q(s) = q^s - 1$. The condition $K_q(s)K_q(1 - s) \in \mathbb{R}$ is algebraically equivalent to $\Re(s) = 1/2$ or $\Im(s) = n\pi/\ln q$, as the same calculation shows. The critical line $\sigma = 1/2$ is therefore universal for all bases q.

5 Discussion

The phase condition $K(s)K(1-s) \in \mathbb{R}$ provides an algebraic characterisation of the critical line $\Re(s) = 1/2$: it is the unique line (aside from the discrete set of exceptional lines) on which the valence function $K(s) = 2^s - 1$ and its reflection $K(1 - s)$ have real product.

The numerical data suggest that no non-trivial zero of $\zeta(s)$ lies on an exceptional line, but a rigorous analytic proof of this fact—and hence a proof that the phase condition is equivalent to the Riemann Hypothesis—remains open.

References

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